



Tablet Deployment Instructions

D.1. Deploying the Remote Health Monitoring System on a New Tablet

D.1.1. Prerequisites

Before starting:

- Connect the tablet to Wi-Fi.
- Ensure sufficient free storage space is available.
- Ensure the tablet is connected to power during deployment.

D.2. Prepare Android Settings

D.2.1. Remove Lock Screen Security

Automatic startup after reboot requires the tablet to unlock automatically.

Open:

Settings → Security

Set the screen lock to "None" or "Swipe"

Do not configure a pin, password or pattern.

Optionally verify that rebooting the tablet does not require user interaction before applications can start.

D.2.2. Disable Battery Optimization

Android battery optimization can prevent Termux and Termux:Boot from running reliably after reboot.

Open:

Settings → Apps → Special Access → Battery Optimization

Locate and disable battery optimization for:

- Termux
- Termux:Boot
- Fully Kiosk Browser

The exact menu structure may vary by Android version and manufacturer.

D.3. Install Required Applications

Install:

1. Termux (from F-Droid)

2. Termux:Boot (from F-Droid)
3. Fully Kiosk Browser (from Play Store)

F-Droid is the recommended app store from which to install Termux, as the version available in the Play Store is not up to date. F-Droid can be installed at: <https://f-droid.org/en/>

D.4. Open Termux:Boot Once

Termux:Boot must be opened at least once after installation, to allow the system to start the application at boot in future.

D.5. Open Termux

Launch Termux and wait for the terminal prompt.

D.6. Install Git

Execute:

```
pkg update -y
```

```
pkg install git -y
```

Wait for completion.

Note: if you see the message "no mirror or mirror group selected", do the following before executing the commands again:

- execute

```
termux-change-repo
```
- select mirror group
- select Europe

D.7. Clone the Repository

Execute:

```
git clone https://github.com/frederikschiepers123/BEP-Remote-Health-Monitoring-System-for-the-Elderly
```

Enter the repository:

```
cd BEP-Remote-Health-Monitoring-System-for-the-Elderly
```

D.8. Make the Deployment Script Executable

Execute:

```
chmod +x deploy.sh
```

D.9. Run the Deployment Script

Execute:

```
./deploy.sh
```

The deployment script will:

- Request storage permissions
- Install Node.js
- Install MagicMirror dependencies
- Install MQTT dependencies

- Install the boot script
- Copy the Fully Kiosk configuration file
- Copy the Android APK
- Launch the APK installer

D.10. Grant Storage Permission

When Android requests storage access:

1. Select Allow.
2. Return to Termux.
3. Press Enter when requested.

The deployment process will continue automatically.

D.11. Step 10 – Install the Android Application

When Android displays the APK installation screen:

1. Select Install.
2. Wait for installation to complete.
3. Open the application once if requested.

D.12. Enable Device Administrator Permission

The Remote Health Monitoring application must be granted Device Administrator privileges.

Open:

Settings → Security → Device Admin Apps

(or the equivalent menu on the device)

Enable:

Remote Health Monitoring Application (called "HealthMonitorWakeTest")

Confirm the permission request.

Note: The exact application name may differ from the APK filename.

D.13. Import Fully Kiosk Configuration

Open Fully Kiosk Browser.

Navigate to:

Settings → Other Settings → Import Settings

Browse to:

Downloads/fully-settings.json

Import the configuration file.

Wait for Fully Kiosk Browser to apply the settings.

D.14. Reboot the Tablet

Restart the tablet.

Wait approximately 30 seconds after startup.

D.15. Verify Automatic Startup

After reboot, verify that:

1. Android boots directly to the home screen.
2. No PIN, password or pattern is requested.
3. Termux:Boot executes successfully.
4. MagicMirror starts automatically.
5. Fully Kiosk Browser launches automatically.
6. The dashboard becomes visible.

D.16. Troubleshooting

D.16.1. MagicMirror did not start

Open Termux and execute:

```
cat ~/magicmirror.log
```

Review any reported errors.

D.16.2. Boot Script Did Not Execute

Execute:

```
cat ~/boot-test.log
```

A recent timestamp should be present.

If no timestamp is present:

- Verify Termux:Boot was opened once after installation.
- Verify battery optimization is disabled.

D.16.3. Fully Kiosk Did Not Start

Verify:

- Fully Kiosk Browser is installed.
- The configuration file was imported successfully.
- Battery optimization is disabled.

D.16.4. Device Stays on Lock Screen After Reboot

Verify:

Settings → Security

Screen lock must be "None" or "Swipe"

Do not use a PIN, password or pattern lock.

D.17. Deployment Complete

Deployment is successful when:

- The custom Android application is installed.
- Device Administrator permission is enabled.
- Fully Kiosk Browser is configured.
- MagicMirror starts automatically after reboot.
- Fully Kiosk Browser launches automatically after reboot.
- The Remote Health Monitoring dashboard is visible.